

CS 365 Midterm Notes

Acquisition Type	Data Format
Web scraping	HTML → Table
API	JSON, XML (hierarchical)
Downloads	CSV, XLSX (flat tables)
Surveys	Forms, Apps (unstructured)

Key Considerations

1. Informed consent
2. Privacy protection
3. Cultural sensitivity
4. Anonymization

Data Source	Use
Public data sets	Prototypes, class projects, benchmarks
Enterprise / Proprietary	critical decision making, organization source of truth
APIs	real-time data feeds ↳ up-to-date, incremental data
Internal, user-generated data	- forms, apps, logs - used for specific data does not exist

Data Governance

Public ≠ Free to Use - always verify the terms of use and licensing agreements

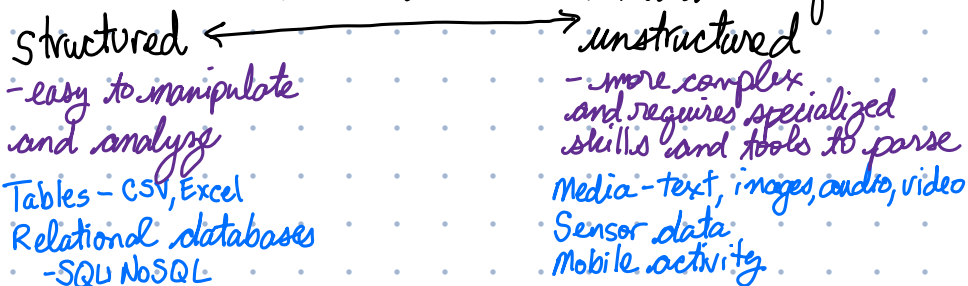
Open Data
freely available,
reusable,
shareable

Proprietary Data
restricted access,
license-based

    commercial Remix Share	No copyright Public domain, attribution required, share-alike
  Remix Share	attribution required, share-alike, non-commercial
 Share	attribution required, non-commercial, no derived works
 all rights reserved	Copyright

Data Wrangling - process of cleaning, transforming, and enriching raw data

Goal: convert structured and unstructured data into a usable format



The ETL Process

Extract - retrieve data from source systems

Transform - clean, standardize, and restructure the data

1. Clean stage - make data "not wrong"

Fixing types, standardization, deduplication, imputation

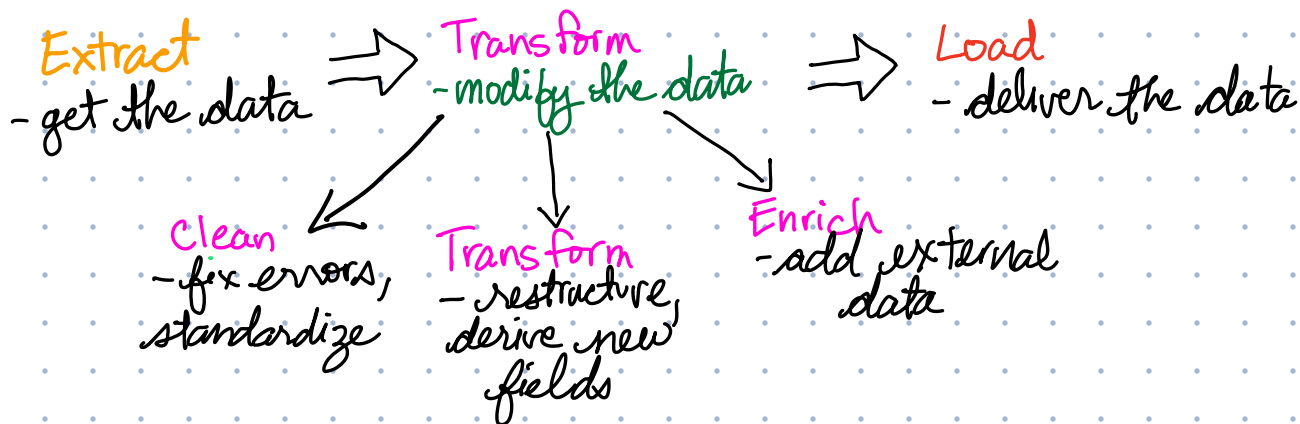
2. Transform stage - make data fit for specific questions

Filter data to relevant time frame, reshaping - combining sets, create new columns, perform calculations

3. Enrich stage - add useful context → application, value

Look up tables, region names, incorporate calendars, add event information

Load - insert processed data into a target system



Feature engineering - creating meaningful features from raw data and helps improve model compatibility

e.g. formatting timestamps for readability

Domain Knowledge + Creativity

Raw Data
- tables
- logs
- timestamps



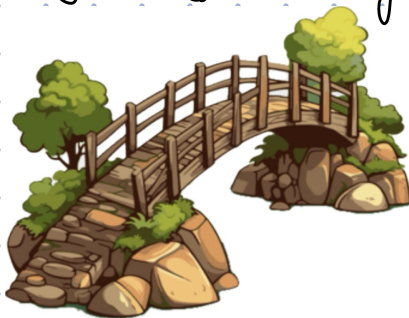
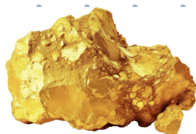
Feature Engineering
- change data type
- standardize
- impute
- bin
- aggregate
- combine
- parse dates



Feature Set & Model
- cleaner feature set

Feature engineering "The creative bridge between raw data and models"

raw data



model



* Feature engineering and data integration are crucial skills that bridge raw data and meaningful analysis, enabling more effective decision-making across various domains.

- transforms raw, messy data into actionable insights
- create a reliable and unified dataset

Types of Features

Numerical - continuous quantitative values

Categorical - discrete labels that are non-numerical

Text-based - data tags, search queries

Temporal - related to time

Data Discovery / Exploration - examining data types, identifying patterns and detecting missing or incomplete data

Transformation / Structuring - reshaping and organizing the data into a suitable format for analysis

Types of Transformations

Normalization and scaling - z-scores

Binning - continuous values into discrete categories

Encoding - categorical data into numerical values

Log transformations & polynomials - handles skewed and non-linear relationships

Reshaping data: pivoting, unpivoting, or transposing data to change its orientation

Pivoting Data

Wide Format (Pivot): one variable per column

- better readability

Long Format (Unpivot): variables stacked in rows

- process of transforming data from a wide format with multiple columns to a long, narrow format, essentially converting columns to rows

- database-friendly

↳ better for analysis, visualization, filtering, grouping



id	GiftCard	T-Shirt	Shipping
1	1	NULL	3
2	2	5	4
3	1	3	10



id	Product Name	Product Qty
1	GiftCard	1
1	Shipping	3
2	GiftCard	2
2	T-Shirt	5
2	Shipping	4
3	GiftCard	1
3	T-Shirt	3
3	Shipping	10

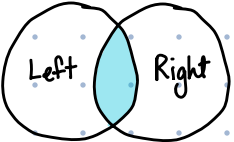
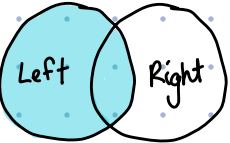
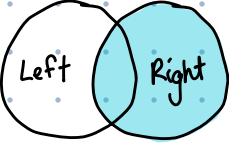
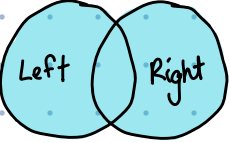
Data Integration - the process of combining data from multiple different kind of sources into a unified, coherent set
e.g. Merging online and in-store transactions

It is NOT data aggregation

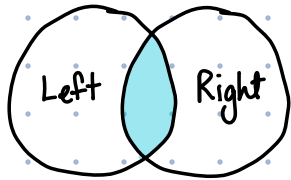
Merging	Integration	Enrichment	combining data from different sources
Aggregating	Summarization Transformation	Enrichment	performing calculations w/ a single table

Schema mapping - align types and standardize names and columns before joining
- harmonize units

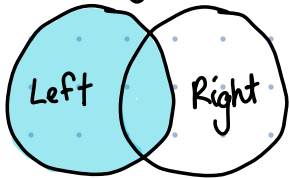
Merge / Joining data: Combining data from multiple sources based on common identifiers

Inner		Overlapping records
Left		primary data with optional enrichment
Right		
Full		all data

Inner join - returns only the rows where the join condition is met in both tables
- finding the perfect match
- stuck in the middle with you.

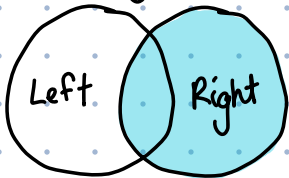


Left join - returns all rows from the left table and only the matching rows of the right table



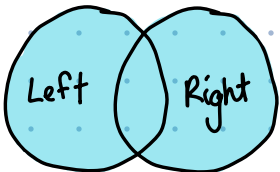
- "To the left, to the left
Everything you own in the box to the left"

Right join - returns all rows from the right table and only the matching rows of the left table

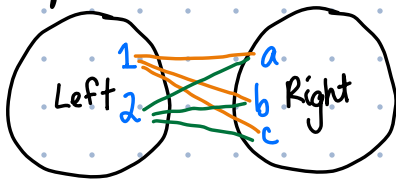


"Right here, right now, there's no other place I wanna be"

Full Outer join - returns all rows when there is a match in either the left or right table - "leave nothing behind"

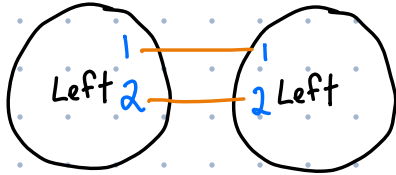


Cross join - returns all possible combinations of rows from the first table with all the possible combinations of rows from the second table



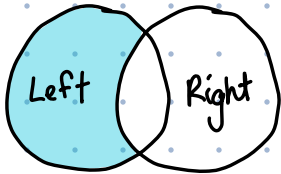
- "The Cartesian Product"

Self join - a table is joined to itself (useful for comparing rows)

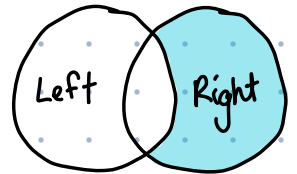


- "A table meets itself"

Left anti-join "Clowns to the left of me"



Right anti-join



"Jokers to the right"

Conflict resolution - conflicting sources of data

Data quality - ensure uniqueness

- verify completeness and reasonableness of values

Validation - verifying the quality, consistency, and accuracy of wrangled data
↳ often by applying validation rules, and checks to ensure data integrity

Publishing - preparing the cleaned and transformed data for downstream use, such as analysis, reporting, or ML and including wrangling documentation

Types of Derived Fields

- calculated
- lookup
- domain logic

Types of Data Integration

ETL - Extract, Transform, Load

ELT - Extract, Load, Transform

- load data into the target
- transform within the target system

EAI - enterprise application integration

- application integration and data sharing in real-time

Key considerations

Data matching - Is the data from different sources the same?

Keys and joins - Do the data sets have primary and foreign keys in common?

Identity resolution - Can different identifiers be linked?

Integration Architecture	Description	Example	Pros	Cons
Centralized	• all the data in one place	Data warehouse	• easy to manage • clean and prepared for analysis	• costly and slow to update
Federated	• data remains in the original location • virtual layer provides a unified view • retrieve and manipulate the data w/o requiring technical details like how it is formatted and stored	Data Virtualization	• reduces data duplication • real-time access	• can be slow for large queries
Real-time	• data integration as it is generated • uses streaming pipelines or APIs	Streaming, APIs	• real-time analytics	• requires a robust infrastructure

Types of Missing Data

1. Missing Completely at Random (MCAR)

- missing values are randomly distributed
- no relationship between missing data

e.g. survey responses lost due to system glitches

Impact - minimal bias if the sample size is adequate

2. Missing at Random (MAR)

- missing data is related to observed variables but not the missing variable itself

e.g. Younger people more likely to skip income questions

Impact: Can be handled with modelling techniques

3. Missing not at Random (MNAR)

- missing data is related to the unobserved value itself

e.g. High earners refusing to disclose income

Impact: Most problematic; requires domain expertise to address

Type	Depends on...	Safe to Delete?	Safe Methods
MCAR	 Completely random - Pure chance	✓ YES	• Like wise deletion • Central tendency imputation • any imputation
MAR	 Observable pattern - other observed variables	⚠ With caution	• predictive imputation • multiple imputation • regression imputation • KNN imputation ⚠ Likewise deletion - will not bias the data set if observed variables are accounted for
MNAR	 Hidden cause - The missing value itself	⊘ No-biased	• Sensitivity analysis • Pattern-mixture models • Selection models • Domain Knowledge estimates ⊘ Simple deletion - introduce bias ⊘ Simple imputation - may mask the systematic bias

Types of Imputation

1. Remove (Deletion)

Listwise deletion - remove rows with missing values

- can maintain integrity, but reduces the sample size
- if data is not MCAR, will introduce bias

Pairwise deletion - use all available for each analysis

2. Fill (Imputation)

Simple imputation

- mean, median or mode substitution
- forward or backward fill for time series
- constant value fill (most often 0)

Advanced imputation

- multiple imputation
- KNN
- regression-based
- ML

3. Flag (Create indicators)

Create boolean columns to indicate missingness

- allows for analysis as to why the data is missing
↳ missingness itself is meaningful sometimes

Use Domain Knowledge

- regulatory or operational constraints
- provide context behind missing values
- document assumptions and rationale for imputation

Outliers - a data point that differs significantly from the other observations, outside the typical range of the dataset

- can distort statistical analysis
- may indicate data entry or measurement errors
- could represent a legitimate, but rare event

INVESTIGATE FIRST. Determine the cause then decide whether to

- Keep them
- Cap them (assign a max or min value)
- Remove them

Standardization & Deduplication

Standardization - making data consistent and uniform across the dataset by

- Trimming
- Adjusting case
- Removing diacritics
- Unifying the formatting

Standardize BEFORE Deduplication

- catch true duplicates that appear differently due to formatting

Benefits

- more reliable equality and matching checks
- fewer false negatives when deduplicating
- better quality data and performance in joins and lookups

Deduplication Best Practice - select multiple columns together to uniquely identify records

e.g. Remove duplicates based on Customer ID, Order Date, and Product ID rather than just one field that may legitimately repeat

ETL	Extract → Transform → Load
Data Integration	Combining data from multiple sources
Ethics	Public data ≠ free to use
Missing Data	Deletion isn't always best
Outliers	Investigate first
Data Types	Choose to preserve formatting
Standardization	BEFORE deduplication
Feature Engineering	Create new variables from existing data

Example from assignment #2

Clean - standardization, deduplication

Transform - normalize the consent & channel categories

Enrich - joining the zip code look up table to impute zip codes
(external referenced data)

Transform - standardize dates, deriving days-between column, adding quality flags for validation

↳ identify issues w/o deleting, enable downstream filtering and analysis, measure data quality metrics, support data lineage and auditability, prioritize data cleaning efforts

Transform - segmentation of age-groups, aggregating data for analysis, prepare data for downstream use